Department of Aeronautical Engineering
Annasaheb Dange College of Engineering & Technology, Ashta

INTERNATIONAL CONFERENCE

Research and Developments in Aeronautical Engineering

RDAE 2027

In association with
AeSI Pune Chapter

Connecting Research, Innovation and Industry for a Sustainable Aerospace Future

12 & 13 March 2027
Two-day international conference

ADCET, Ashta
Dist. Sangli, Maharashtra, India

About RDAE 2027

The International Conference on Research and Developments in Aeronautical Engineering brings together researchers, academicians, industry professionals and students to share knowledge, present innovations and discuss emerging trends and challenges in aeronautical engineering. It offers a platform for collaboration and exchange of ideas to accelerate research, foster innovation and contribute to a sustainable aerospace future.

6 Tracks
Across aerospace disciplines

Peer Review
Managed via Microsoft CMT

Keynotes
From industry & AeSI

15 Dec 2026
Paper submission deadline

CONFERENCE WEBSITE
rdae2027.adcet.ac.in

CONTACT
rdae@adcet.ac.in

CALL FOR PAPERS

Conference Tracks & Topics

Original research papers, review papers, case studies and industry presentations are invited under the following six tracks.

TRACK 01
Aerodynamics, Fluid Mechanics & CFD
Track Chair: Dr. Kaleeswaran P.

- Subsonic to hypersonic aerodynamics
- Airfoil & wing design and optimisation
- Boundary layer, transition & turbulence modelling
- Active & passive flow control, vortex generators
- High-lift devices & low-Reynolds-number flows
- Unsteady aerodynamics & aeroacoustics
- CFD methods, meshing & solver development
- CFD verification, validation & uncertainty
- Wind tunnel testing & flow diagnostics
- Heat transfer & thermal management
- Multiphase, reacting & compressible flows
- Fluid–structure interaction & aeroelasticity

TRACK 02
Structures
Track Chair: Mr. Oblisamy L.

- Aircraft & spacecraft structural analysis and design
- Finite element & computational structural mechanics
- Fatigue, fracture & damage tolerance
- Structural dynamics, vibration & modal analysis
- Buckling & stability of thin-walled structures
- Impact, crashworthiness & blast resistance
- Structural health monitoring & NDE
- Lightweight design & topology optimisation
- Aeroelasticity, flutter & gust response
- Joints, fasteners & bonded repairs
- Thermal & thermo-mechanical stress analysis
- Reliability, life prediction & airworthiness

TRACK 03
Composites & Advanced Materials
Track Chair: Mr. Basithrahman A.

- Fibre-reinforced polymer composites
- Metal matrix & ceramic matrix composites
- Nanocomposites & nanomaterials
- Smart, functional & shape-memory materials
- Lightweight alloys & superalloys
- Additive manufacturing & 3D printing
- Composite manufacturing & process modelling
- Mechanical, thermal & microstructural characterisation
- Damage, delamination & failure of composites
- High-temperature & thermal protection materials
- Coatings, corrosion & surface engineering
- Bio-based, recyclable & sustainable materials

TRACK 04
Aerospace Propulsion & Power Systems
Track Chair: Mr. Arun Nema

- Gas turbine engines: compressors, combustors, turbines
- Combustion, emissions & sustainable aviation fuels
- Solid, liquid & hybrid rocket propulsion
- Ramjet, scramjet & hypersonic propulsion
- Electric, hybrid-electric & hydrogen propulsion
- Propeller, rotor & fan design and performance
- Turbomachinery aerodynamics & blade cooling
- Nozzles, inlets & thrust vectoring
- Batteries, fuel cells & energy storage
- Engine health monitoring & performance testing
- Electric propulsion for spacecraft
- Aircraft power, thermal & environmental control

TRACK 05
UAVs & Autonomous Aerial Systems
Track Chair: Mr. Reju R.

- Fixed-wing, rotary-wing & hybrid UAV design
- UAV flight dynamics, stability & control
- Guidance, navigation & control (GNC)
- Path planning & obstacle avoidance
- Swarm & multi-agent UAV systems
- Sensors, avionics & data links
- eVTOL, urban air mobility & drone delivery
- UAVs for agriculture, mapping & disaster response
- Micro air vehicles & bio-inspired flight
- Hardware-in-the-loop simulation & flight testing
- Airspace integration, regulation & safety
- Solar, hybrid & long-endurance UAVs

TRACK 06
AI/ML & Intelligent Aerospace Systems
Track Chair: Dr. Sanoj P. Suresh

- Machine learning & deep learning in aerospace
- AI-driven aerodynamic & structural optimisation
- Physics-informed neural networks & surrogates
- Predictive maintenance & health management
- Digital twins of aircraft, engines & factories
- Computer vision for inspection & navigation
- Reinforcement learning for flight control
- Intelligent avionics & pilot assistance
- AI in air traffic management & operations
- Analytics of flight, sensor & test data
- Explainable, safe & certifiable AI for aviation
- Cybersecurity of intelligent aerospace systems

Topics listed are indicative, not exhaustive. Papers on any other relevant topic within a track, or in allied areas of aeronautical engineering, are equally welcome.

MARK YOUR CALENDAR

Conference Important Dates

Call for Papers	15 Oct 2026
Paper Submission Deadline	15 Dec 2026
Paper Review Communication	15 Jan 2027
Revised Paper Submission (accepted papers)	25 Jan 2027
Final Acceptance Communication	07 Feb 2027
Early Bird Registration	15 Feb 2027
Submission of Camera-Ready Paper	20 Feb 2027
Regular Registration	11 Mar 2027
Conference	**12 & 13 March 2027**

PAPER SUBMISSION

How to Submit

1. **Prepare** your full paper using the RDAE 2027 paper template (download from the website).
2. **Create** and verify an account at cmt3.research.microsoft.com.
3. **Find the conference:** under "All Conferences", search for **RDAE2027**.
4. **Create a new submission**, enter author details and upload your full paper.
5. **Track** reviews and upload the revised and camera-ready versions in CMT.

cmt3.research.microsoft.com/RDAE2027

Research papers · Review papers · Case studies · Industry presentations

REGISTRATION

Registration Fees

Category	Early Bird (Till 15th Feb 2027)		Regular (After 15th Feb 2027)	
	National (INR)	International (USD)	National (INR)	International (USD)
UG Students	₹1,000	$40	₹1,500	$50
PG Students & Research Scholars	₹1,500	$60	₹2,000	$75
Academicians	₹2,500	$100	₹3,000	$125
Industry Participants	₹4,000	$150	₹5,000	$200
Accompanying Person	₹700	$30	₹1,000	$40

Publication charges are additional and will be conveyed soon.

INR fees apply to participants from India; USD fees to participants from outside India.

Keynote & Invited Speakers

Keynote speakers will be eminent experts from the aerospace industry and the Aeronautical Society of India (AeSI). Details will be updated on the website soon.

ORGANISING TEAM

Conference Committee

Chief Patrons: Sant Dnyaneshwar Shikshan Sanstha, Urun-Ishwarpur (formerly known as Islampur)



Chief Patron
Hon. Annasaheb Dange
Founder & Chairman, SDSS



Chief Patron
Adv. Rajendra R. Dange
Secretary, SDSS



Chief Patron
Hon. Vishwanath R. Dange
Joint Secretary, SDSS



Patron
Dr. Laxman Y. Waghmode
Director, ADCET



Advisory Committee Chair
Dr. Amol S. Dange
Dean, RIS, ADCET



Conference Chair
Dr. Yuvaraj S.
HoD, Aeronautical Engg.



Convenor
Dr. S. Sendhil Kumar
Professor, Aero. Engg.



Co-Convenor
Mr. Yogesh B. Kumbhar
Asst. Prof., Aero. Engg.

Track Chairs

Aerodynamics, Fluid Mech. & CFD **Dr. Kaleeswaran P.**	Structures **Mr. Oblisamy L.**	Composites & Advanced Materials **Mr. Basithrahman A.**
Propulsion & Power Systems **Mr. Arun Nema**	UAVs & Autonomous Aerial Systems **Mr. Reju R.**	AI/ML & Intelligent Aerospace Systems **Dr. Sanoj P. Suresh**

National Advisory Committee

Dr. N. Kishore Nath
DRDO, Hyderabad

Anandaraj A.
Scientist, ARDE · Hon. Secretary (General), AeSI Pune Chapter

Dr. A. P. Senthil Kumar
PSG College of Technology, Coimbatore

Dr. K. Sundararaj
Kumaraguru College of Technology

Dr. P. Manivannan
Hindustan University, Chennai

Dr. C. J. Thomas Renald
SREC, Coimbatore

Dr. M. Manikandan
MIT, Manipal

Dr. M. Sathesh
IIT Kanpur

Dr. S. Janaki
SNS College of Engineering, Coimbatore

Mr. A. K. L. Bhagat
Jt. Director, DRDO · Jt. Secretary (PR), AeSI Pune Chapter

Mr. Sunil B. Hiwale
DRDL, Hyderabad · ARDE

Dr. Karthikeyan P.
PSG College of Technology, Coimbatore

Dr. K. Jayabal
IIITDM, Kancheepuram

Dr. Raguraman Munusamy
IIITDM, Kancheepuram

Dr. P. S. Prem Kumar
Kumaraguru College of Technology

Dr. Devabrata Sahoo
MIT ADT University

Dr. N. Mathan Kumar
KPRIET, Coimbatore

Dr. L. K. Gite
Sr. Defence Scientist, ARDE · Jt. Secretary (PR), AeSI Pune Chapter

Dr. M. Senthilkumar
PSG College of Technology, Coimbatore

Dr. T. V. Arjunan
Guru Ghasidas Vishwavidyalaya, Bilaspur

Dr. V. Satheesh Kumar
NIT, Tiruchirappalli

Dr. Santhana Krishnan
CMR University, Bengaluru

Dr. B. N. Sreeharan
Kumaraguru College of Technology

Dr. G. Prasad
Chandigarh University, Punjab

Dr. Muthukumar M.
Nandha Engineering College, Erode

VENUE & HOW TO REACH

Annasaheb Dange College of Engineering & Technology, Ashta, Tal. Walwa, Dist. Sangli, Maharashtra – 416 301

Rail: Sangli / Miraj Jn. · Air: Kolhapur (KLH) or Pune (PNQ) · Road via Urun-Ishwarpur / Sangli

CONTACT THE CONVENOR

rdae@adcet.ac.in

rdae2027.adcet.ac.in